

1 Curves in $\mathbb{R}^3$

1.1 Basic Definitions of curves

- **Length of a Curve:** The length of a parameterized curve c is $L(c) := \int_a^b \|\dot{c}(t)\| dt$.
- **Regular Curve:** A curve $c(t)$ is regular if $\dot{c}(t) \neq 0$ for all $t \in I$.
- **Arc Length Parametrization:** A curve $c(s)$ is unit-speed if $\|\dot{c}(s)\| = 1$.

For every regular curve, $\psi(t) = \int_{t_0}^t \|\dot{c}(t)\| dt, \psi^{-1} \equiv \varphi, \tilde{c} = c \circ \varphi$ is unit-speed.

- **Frenet Frame:** If c is a unit-speed curve, $T(t) := \dot{c}(t), N(t) := \frac{\ddot{c}(t)}{\|\ddot{c}(t)\|}$ has unit length and are perpendicular to each other.

- **Curvature (κ):** Measures how fast the curve turns. For a unit-speed curve, $\kappa(t) = \|\ddot{c}(t)\|$.

Generally, $\kappa(t) = \frac{\|\dot{c}(t) \times \ddot{c}(t)\|}{\|\dot{c}(t)\|^3} = \frac{\sqrt{\|\ddot{c}(t)\|^2 - \dot{v}(t)^2}}{v(t)^2} = \frac{\sqrt{v(t)^2 \|\ddot{c}(t)\|^2 - (\dot{c}(t) \cdot \ddot{c}(t))^2}}{v(t)^3}$

- **Torsion (τ):** Measures how much the curve twists out of the plane of curvature.

For a general curve, $\tau(t) = \frac{(\dot{c}(t) \times \ddot{c}(t)) \cdot \ddot{\ddot{c}}(t)}{\|\dot{c}(t) \times \ddot{c}(t)\|^2}$.

- **Frenet-Serret Frame:** Let c be a unit-speed curve in $\mathbb{R}^3$, $T(t) = \dot{c}(t), N(t) = \frac{\ddot{c}(t)}{\|\ddot{c}(t)\|} =$

$\frac{\ddot{c}(t)}{\|\kappa(t)\|}, B(t) = T(t) \times N(t)$ are the tangent field, the normal field and the binormal field. $\{T, N, B\}$ is called the Frenet-Serret frame.

- **Frenet-Serret Formulas:**
$$\begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

1.2 Extra Applications

- **Isoperimetric Inequality:** For a closed curve of length L enclosing area A , $L^2 \geq 4\pi A$ with equality iff the curve is a circle.
- **Winding Number:** Let c a loop and P_0 be a point not lying on the curve, the winding number $W(c, P_0)$ is the number of times c goes counter clockwise around P_0 .

If $c(t) = P_0 + r(t)(\cos \theta(t), \sin \theta(t))$ and $P_0 = 0$, then $W(c, P_0) = \frac{1}{2\pi} \int_a^b \frac{\dot{y}x - \dot{x}y}{x^2 + y^2}(t) dt$.

1.3 Calculation Examples

For a plane curve $c(t) = (x(t), y(t))$, the curvature is given by: $\kappa = \frac{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|}{(\dot{x}^2 + \dot{y}^2)^{3/2}}$.

2 Surfaces in $\mathbb{R}^3$

2.1 Definition and Construction of Surfaces

- **Regular Map:** A map Φ is called regular if $\frac{\partial \Phi}{\partial x}(p)$ and $\frac{\partial \Phi}{\partial y}(p)$ are not zero and not parallel.

- **Coordinate Chart:** Given a subset $S \subseteq \mathbb{R}^3$, a coordinate chart on S around a point $m \in S$, consists of
 - an open subset $D \subseteq \mathbb{R}^2$
 - a regular, injective smooth map $\Phi : D \rightarrow S$, $\Phi(D) \subseteq S$
 - an open subset $\theta \in \mathbb{R}^3$, $m \in \theta$

such that $\Phi(D) = S \cap \theta$ and $\Phi^{-1} : \theta \cap S \rightarrow D$ is continuous.

- **Surface:** A subset $S \subseteq \mathbb{R}^3$ is a surface if every point $m \in S$ has a coordinate chart centered at m .

There are three common surfaces:

- **Ruled Surface:** A ruled surface is a surface swept only by straight line $l : I \rightarrow \mathbb{R}^3$ moving along some curve $c : I \rightarrow \mathbb{R}^3 \Rightarrow \Phi(t, r) = c(t) + rl(t), r \in \mathbb{R}$
- **Graphs:** Every smooth function of $D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ determines a parameterized surface $S = \text{graph } f = \{(x, y, z) \in D \times \mathbb{R} : z = f(x, y)\}$
- **Level Sets:** Let $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ a smooth function. For every $c \in \mathbb{R}$, consider $S = g^{-1}(C) = \{(x, y, z) \in \mathbb{R}^3, g(x, y, z) = c\}$ is called a level set.
If $\left(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z}\right) \neq 0$ on $g^{-1}(C)$, then S is a surface.
- **Smooth Map:** Given a surface S , a function $f : S \rightarrow \mathbb{R}^3$ is called smooth if all compositions $f \circ \Phi : \mathbb{R}^2 \supseteq D \rightarrow \mathbb{R}^3$ are smooth for every coordinate chart $\Phi : D \rightarrow S$.

2.2 Vector Fields and Tangent Space

- **Tangent Vector:** A tangent vector X at $m \in S$ is $X \in \mathbb{R}^3$ such that there exists a smooth curve $c : (-\epsilon, \epsilon) \rightarrow S$ with $c(0) = m$ and $\dot{c}(0) = X$.
- **Tangent Plane:** The set of all tangent vectors at $m \in S$ is called the tangent plane at m , denoted by $T_m S$.

Cal Method: Take the derivative of the surface ∇F , and the tangent plane is

$$\{\omega \in \mathbb{R}^3 : \nabla F \cdot \omega = 0\}$$

- **Vector Field:** A vector field on S is a map $X : S \rightarrow \mathbb{R}^3$ such that $X(m) \in T_m S$ for all $m \in S$.
- **Directional Derivative:** Directional derivative of $f : S \rightarrow \mathbb{R}^k$ in the direction $X \in T_m S$ is defined by $T_m f(X) = \left. \frac{d}{dt} \right|_t f(m + tX)$.
- **Vector field under local parametrization:** Given a coordinate chart $\Phi(u, v) : D \rightarrow S$, a tangent vector X on S is the span of $\partial_u \Phi, \partial_v \Phi$.

The tangent plane at $m = \Phi(u_0, v_0)$ is $T_m \Phi(x, y) = x \partial_u^\Phi(m) + y \partial_v^\Phi(m), (x, y) \in \mathbb{R}^2$.

2.3 Differential Forms

- **1-Form on Surface:** A 1-form on S is a linear map $\alpha_m : T_m S \rightarrow \mathbb{R}$ for each point $m \in S$. We write α for the whole family. $\alpha_m(v) = \alpha_m(v_1 \partial_x + v_2 \partial_y) = v_1 \alpha_m(\partial_x) + v_2 \alpha_m(\partial_y)$.

1-form is called closed if $d\alpha = 0$ and is called exact if there exists a smooth function $f : S \rightarrow \mathbb{R}$ such that $\alpha = df$. exact implies closed.

- **1-Form** $(dx)_p, (dy)_p$: 1-form $(dx)_p$ is defined by $(dx)_p(\partial x)_p = 1, (dx)_p(\partial y)_p = 0$ and similarly for $(dy)_p$. Moreover, every 1-form α at p can be expressed as $\alpha = \alpha_1(p)(dx)_p + \alpha_2(p)(dy)_p$, α_1, α_2 are smooth functions.
- $T_p f$: Given a smooth map $f : S \rightarrow \mathbb{R}$, the directional derivative $T_p f$ is a 1-form at p defined by $(df)_p(v) = T_p f(v) = (\nabla f) \cdot v$.
- **Local Representation of 1-Form**: Given a coordinate chart $\Phi(u, v) : D \rightarrow S$, any 1-form α on D , then $\alpha_{(x,y)}^\Phi = \alpha_{\Phi(x,y)} \circ T_{(x,y)} \Phi$.
Cal Method: If $\alpha = \alpha_1 dx + \alpha_2 dy$, then $\alpha^\Phi = \alpha_1(\Phi(u, v)) dx^\Phi + \alpha_2(\Phi(u, v)) dy^\Phi$.
- **Integration of 1-Form along Curve**: Given a smooth curve $c : [a, b] \rightarrow S$ and a 1-form α on S , the integral of α along c is defined by $\int_c \alpha := \int_a^b \alpha_{c(t)}(\dot{c}(t)) dt$.
Specialt, if $\alpha = df$ and f is a smooth function, then $\int_c df = f(c(b)) - f(c(a))$.
- **2-Form**: A 2-form β on S is, for each point $m \in S$, a bilinear antisymmetric map $\beta_m : T_m S \times T_m S \rightarrow \mathbb{R}$. In particular, $\beta_m(v, v) = 0$.
- **k-Form**: A k-form ω on S is, for each point $m \in S$, a multilinear antisymmetric map $\omega_m : (T_m S)^k \rightarrow \mathbb{R}$.
- **Wedge Product**: Given two 1-form α_1, α_2 , their wedge product $\alpha_1 \wedge \alpha_2$ is a 2-form defined by $(\alpha_1 \wedge \alpha_2)_m(u, v) = (\alpha_1)_m(u)(\alpha_2)_m(v) - (\alpha_1)_m(v)(\alpha_2)_m(u), (u, v) \in T_m S$.
Note that $\alpha_1 \wedge \alpha_2 = -\alpha_2 \wedge \alpha_1$.
- **Exterior Derivative of 1-Form**: α is a 1-form on S , its exterior derivative $d\alpha$ is a 2-form defined by $(d\alpha)_m(\partial_x^\Phi(m), \partial_y^\Phi(m)) = T_m(\alpha(\partial_y^\Phi))(\partial_x^\Phi(m)) - T_m(\alpha(\partial_x^\Phi))(\partial_y^\Phi(m))$ for a chart $\Phi : D \rightarrow S$.
Cal Method: If $\alpha = \alpha_1 du + \alpha_2 dv$, then $d\alpha = \left(\frac{\partial \alpha_2}{\partial u} - \frac{\partial \alpha_1}{\partial v} \right) du \wedge dv$.
- **Properties of Exterior Derivative**: $f : S \rightarrow \mathbb{R}, \alpha$ 1-form, $d(f\alpha) = df \wedge \alpha + f d\alpha, d(df) = 0$

2.4 Cartan's Moving Frame

- **Moving Frame**: A Moving frame on an open subset $U \subseteq \mathbb{R}^3$ consists of vector fields $e_1, e_2, e_3 : U \rightarrow \mathbb{R}^3$ such that for every $x \in U$, $e_1(x), e_2(x), e_3(x)$ are an orthonormal basis of $\mathbb{R}^3$.
- **Connection 1-Form**: Given a moving frame e_1, e_2, e_3 on S , the connection 1-form ω_i^j is defined by $(de_i)_x(v) = \sum_{j=1}^3 (\omega_i^j)_x(v) e_j(x)$. In short, $de_i = \sum_{j=1}^3 \omega_i^j e_j$. For ONB, $(\omega_i^j)_x(v) = (de_i)_x(v) \cdot e_j(x)$.
- **Connection Form and Frenet-Serret Frame**: For a unit-speed curve c with $\kappa > 0$. Suppose e_1, e_2, e_3 is an moving frame on $\mathbb{R}^3$ such that the restrictions of the fields to c give the Frenet-Serret frame T, N, B . Then $(\omega_1^2)_{c(t)}(\dot{c}(t)) = \kappa, (\omega_1^3)_{c(t)}(\dot{c}(t)) = 0, (\omega_2^3)_{c(t)}(\dot{c}(t)) = \tau$.

- **Dual Frame:** Dual frames are 1-forms ω^k defined by $\omega^k(e_j) = \delta_j^k$.

Cal Method: Given a chart $\Phi(u, v, w)$, then $\begin{cases} dx = \sum_{i=1}^3 \omega^i e_i \\ dx = \frac{\partial \Phi}{\partial u} du + \frac{\partial \Phi}{\partial v} dv + \frac{\partial \Phi}{\partial w} dw \end{cases}$, so we can solve ω^i .

- **First Cartan's Structure Equation:** $d\omega^k + \sum_i \omega_i^k \wedge \omega^i = 0$.
- **Second Cartan's Structure Equation:** $d\omega_i^j + \sum_k \omega_k^j \wedge \omega_i^k = 0$.

2.5 Adapted Frame

- **Adapted Frame:** A moving frame $e_1, e_2, e_3 : \mathcal{O} \rightarrow \mathbb{R}^3$ is called adapted to the surface S if for every $p \in S \cap \mathcal{O}$, $e_3(p)$ is normal to S , i.e. $e_3(p) \times X = 0$ for all $X \in T_p S$.

In particular, if S is oriented and n is a unit normal field on S , then $n(p) = \pm e_3(p)$ for all $p \in S \cap \mathcal{O}$. $e_1(p), e_2(p)$ are a basis of $T_p(S)$, $\forall p \in S \cap \mathcal{O}$.

Cal Method: If $\Phi(u, v)$ is the parametrization of S , $e_1 = \frac{\Phi_u}{\|\Phi_u\|}$, $e_3 = \frac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}$, $e_2 = e_3 \times e_1$.

For adapted frame, $\omega^3 = 0$ on S .

- **Area Element under Adapted Frame:** If e_1, e_2, e_3 is an adapted frame on S , then the area element is given by $dA = \pm \omega^1 \wedge \omega^2$.
- **Positive:** An adapted frame is called positively oriented if $dA = \omega^1 \wedge \omega^2 \iff n = e_1 \times e_2$.

2.6 Gauss Map and Shape Operator

- **Gauss Map:** If $n(p)$ has unit norm for $\forall p \in S$, then we call $n : S \rightarrow S^2 \subseteq \mathbb{R}^3$ the Gauss map of S . Moreover, for the surface $\Phi : D \rightarrow S$, Gauss map n can be written as

$$n(u, v) = \frac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}.$$

- **Shape Operator (Weingarten Map):** The shape operator at $p \in S$ is the linear map $\mathcal{S}_p : T_p S \rightarrow T_p S$ defined by $\mathcal{S}_p(X) = -T_p n(X)$ for all $X \in T_p S$.

If e_1, e_2, e_3 is an adapted frame on S , then $\mathcal{S}_p(v) = (\omega_1^3)_p(v)e_1(p) + (\omega_2^3)_p(v)e_2(p)$.

Cal Method: If $\Phi(u, v)$ is the parametrization of S , then $\mathcal{S}_p(\Phi_u) = -n_u$, $\mathcal{S}_p(\Phi_v) = -n_v$.

2.7 First Fundamental Form and Second Fundamental Form

- **First Fundamental Form:** The first fundamental form at $p \in S$ is the bilinear map $I_p : T_p S \times T_p S \rightarrow \mathbb{R}$ defined by $I_p(v, w) = \langle v, w \rangle$ for all $v, w \in T_p S$.

- **Second Fundamental Form:** The second fundamental form at $p \in S$ is the bilinear map $II_p : T_p S \times T_p S \rightarrow \mathbb{R}$ defined by $II_p(v, w) = \langle v, \mathcal{S}_p(w) \rangle = I_p(v, \mathcal{S}_p(w))$ for all $v, w \in T_p S$.

- **Fundamental Forms under Local Parametrization:** Given a coordinate chart $\Phi(u, v) : D \rightarrow S$, $E = \langle \Phi_u, \Phi_u \rangle$, $F = \langle \Phi_u, \Phi_v \rangle$, $G = \langle \Phi_v, \Phi_v \rangle$, $n = \frac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}$, $e = \langle \Phi_{uu}, n \rangle$, $f = \langle \Phi_{uv}, n \rangle$, $g = \langle \Phi_{vv}, n \rangle$.

Then the first and second fundamental forms at m are given by $I^\Phi = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$, $II^\Phi =$

$$\begin{pmatrix} e & f \\ f & g \end{pmatrix}, \mathcal{S} = II \times I^{-1} = \frac{1}{EG - F^2} \begin{pmatrix} e & f \\ f & g \end{pmatrix} \begin{pmatrix} G & -F \\ -F & E \end{pmatrix}.$$

- **Area Element under Local Parametrization 2:** $dA = \sqrt{EG - F^2} du dv$.

2.8 Curvature

- **Normal Curvature:** The normal curvature in the u -direction is $\kappa_p(u) = II_p(u, u) = \langle u, \mathcal{S}_p(u) \rangle$.
- **Principal Directions and Curvatures:** Principal curvature of S at p is the maximum and minimum values of $\kappa_p(u)$ for $\|u\| = 1, u \in T_p S$. The corresponding directions are called principal directions.

The principal curvatures and directions are the eigenvalues and eigenvectors of the shape operator $\mathcal{S}_p$.

- **Gaussian Curvature:** $K = k_1 k_2$. It is the determinant of the shape operator.
- **Mean Curvature:** $H = \frac{k_1 + k_2}{2}$.
- **Curvatures under Local Parametrization:** $K = \det(S) = \frac{eg - f^2}{EG - F^2}$, $H = \frac{1}{2} \text{tr}(S) = \frac{eG - 2fF + gE}{2(EG - F^2)}$. The normal curvatures in direction $w = a\Phi_u + b\Phi_v$ is given by $\kappa\left(\frac{w}{\|w\|}\right) = \frac{ea^2 + 2fab + gb^2}{Ea^2 + 2Fab + Gb^2}$. Use K and H to find principal curvatures.
- **Umbilic Point:** A point $p \in S$ is called an umbilic point if $\kappa_1(p) = \kappa_2(p)$. Otherwise, $\kappa_1(p)k_2(p) > 0$ is called elliptic point, $\kappa_1(p)k_2(p) < 0$ is called hyperbolic point, and $\kappa_1(p)k_2(p) = 0$ is called parabolic point.
- **Darboux Frame:** An adapted frame on $\mathbb{R}^3$ is called a Darboux frame if $e_1(p), e_2(p)$ are the principal directions of S at p .
- **A Line of Curvature:** A unit-speed curve is called a line of curvature if $\dot{c}(t)$ is always a principal direction of S at $c(t)$. **Cal Method:** Φ is the parametrization of S , then a regular curve $c(t) = \Phi(x(t), y(t))$ is a line of curvature iff $(fE - eF)\dot{x}^2 + (gE - eG)\dot{x}\dot{y} + (gF - fG)\dot{y}^2 = 0$.

2.9 Some equations of frames and curvatures

For an adapted frame e_1, e_2, e_3 and its connection forms and dual frames, we have

- $d\omega^1 = \omega_1^2 \wedge \omega^2, d\omega^2 = \omega_2^1 \wedge \omega^1$
- $\omega_1^3 \wedge \omega^1 + \omega_2^3 \wedge \omega^2 = 0$
- $\omega_1^3 \wedge \omega^2 + \omega^1 \wedge \omega_2^3 = 2H\omega^1 \wedge \omega^2$
- $d\omega_1^2 = -K\omega^1 \wedge \omega^2, d\omega_1^3 = \omega_1^2 \wedge \omega_2^3, d\omega_2^3 = \omega_2^1 \wedge \omega_1^3$
- $\omega_1^3 = II_{11}\omega^1 + II_{12}\omega^2, \omega_2^3 = II_{21}\omega^1 + II_{22}\omega^2$

For Darboux frame, we have

- $d\kappa_1(e_2) = (\kappa_1 - \kappa_2)\omega_1^2(e_1), d\kappa_2(e_1) = (\kappa_1 - \kappa_2)\omega_1^2(e_2)$

2.10 Gauss Curvature and Theorema Egregium

- **Push Forward and Full back:** Let $F : \mathbb{R}^2 \supseteq S \rightarrow \bar{S} \subseteq \mathbb{R}^3$
 - **Push Forward of Curves:** For a curve $c : I \rightarrow S$, the push forward of c by F is $F_*c = F \circ c : I \rightarrow \bar{S}$.
 - **Push Forward of Vectors:** For a vector $v \in T_pS$, $c(t) : I \rightarrow S$ s.t. $\dot{c}(0) = v$, the push forward of X by F is $F_*v = (F \circ c)'(0) = T_pF(\dot{c}(0))$, it's just the derivative $T_pF : T_pS \rightarrow T_{F(p)}\bar{S}$.
 - **Pull Back of functions:** For a function $f : \bar{S} \rightarrow \mathbb{R}$, the pull back of f by F is $F^*f = f \circ F : S \rightarrow \mathbb{R}$.
 - **Pull Back of 1-Forms:** For a 1-form α on $\bar{S}$, the pull back of α by F is $(F^*\alpha)_p(v) = \alpha_{F(p)}(T_pF(v))$ for all $p \in S, v \in T_pS$.
 - **Pull Back of 2-Forms:** The pull back of 2-form β by F is $(F^*\beta)_p(u, v) = \beta_{F(p)}(T_pF(u), T_pF(v))$ for all $p \in S, u, v \in T_pS$.
 - **Propositions:** If α, β are 1-forms on $\bar{S}$, then $F^*(\alpha \wedge \beta) = (F^*\alpha) \wedge (F^*\beta), F^*(d\alpha) = d(F^*\alpha)$.
- **Isometry:** A bijective map $F : S \rightarrow \bar{S}$ is called an isometry if F preserves the inner product of tangent vectors $\langle v, w \rangle = \langle T_pF(v), T_pF(w) \rangle, \forall v, w \in T_pS$, i.e. $F^*\bar{I} = I$.
- **Local Isometry:** A map $F : S \rightarrow \bar{S}$ is called a local isometry if for every $p \in S$, there exists an open neighborhood $\mathcal{O} \subseteq S$ of p such that the restriction $F|_{\mathcal{O}} : \mathcal{O} \rightarrow F(\mathcal{O})$ is an isometry.
- **Distance on Surface:** For $p, q \in S$, the distance between p and q on S is defined by $d(p, q) = \inf \{L(c)\}, c(0) = p, c(1) = q$.
- **Isometry preserves distance:** If $F : S \rightarrow \bar{S}$ is an isometry, then $d(p, q) = \bar{d}(F(p), F(q))$ for all $p, q \in S$.
- **Theorema Egregium:** The Gauss curvature K only depends on the first fundamental form I on S and it's invariant under local isometries $F : S \rightarrow \bar{S}$, i.e. $K(p) = \bar{K}(F(p))$ for all $p \in S$.
- **Behaviour of frame under isometry:** If $F : S \rightarrow \bar{S}$ is an isometry, and e_1, e_2, e_3 is an adapted frame on S , then $\bar{e}_i = F_*e_i = T_pF(e_i(p))$ is an adapted frame on $\bar{S}$. Dual frame $\{\bar{\omega}^1, \bar{\omega}^2\}$ is uniquely determined by $F^*\bar{\omega}^i = \omega^i$ for $i = 1, 2$. $\bar{\omega}_1^2$ is uniquely determined by $F^*\bar{\omega}_1^2 = \omega_1^2$.
- **Surface with all points umbilic:** Let S be a connected surface such that all points are umbilic, S has constant Gaussian curvature $K \geq 0$. And if $K = 0$, then S is a part of plane. If $K > 0$, then S is a part of sphere.
- **Constant Gaussian Curvature Surface:** Let S be a compact surface in $\mathbb{R}^3$ with constant

Gaussian curvature $K > 0$, then S is a sphere of radius $\frac{1}{\sqrt{K}}$.

2.11 Geodesics

- **Straight Line in Plane:** A curve $c(t)$ is a straight line $\iff \ddot{c}(t) = 0 \iff \dot{c}(t)$ are parallel $\iff c$ is the shortest path between two points.
- **Geodesic Curvature:** Let $\{e_1, e_2, e_3\}$ be an adapted frame such that $e_1(c(t)) = \frac{\dot{c}(t)}{v}$, i.e. e_1 is the unit tangent vector of c , e_3 is the unit normal vector of S , $e_2 = e_1 \times e_3$, then the geodesic curvature of c at $c(t)$ is defined by $\kappa_g(t) = \omega_1^2 \left(\frac{\dot{c}(t)}{v} \right) = \omega_1^2(e_1)$.
Other Cal Method: If n is the unit normal vector of S , $r(t)$ is the parametrization of c , then $\kappa_g(t) = \frac{\det(\dot{r}(t), \ddot{r}(t), n(r(t)))}{\|\dot{r}(t)\|^3} = \frac{(\dot{r}(t) \times \ddot{r}(t)) \cdot n(r(t))}{\|\dot{r}(t)\|^3}$.
- **The relation between curvatures:** Let $\kappa(t)$ be the curvature of c at $c(t)$ in $\mathbb{R}^3$, and $\kappa_n(t)$ be the normal curvature of S at $c(t)$ in the direction of $\dot{c}(t)$, then $\kappa^2(t) = \kappa_g^2(t) + \kappa_n^2(t)$ and $\kappa_n(t) = \omega_1^3 \left(\frac{\dot{c}(t)}{v} \right) = \omega_1^3(e_1) = \frac{c'' \cdot n}{\|c'\|^2}$.
- **Geodesic:** A curve c on S is called a geodesic $\iff \kappa_g(t) = 0 \iff \ddot{c}(t)$ is in the normal direction $\iff \omega_1^2(\dot{c}(t)) = 0 \iff c$ is locally the shortest path between two points on S .
Cal Method: If $\Phi(u, v)$ is the parametrization of S , then a regular curve $c(t) = \Phi(x(t), y(t))$ is a geodesic iff $\ddot{x} + \frac{1}{2E}(E_u \dot{x}^2 + 2E_v \dot{x}\dot{y} - G_u \dot{y}^2) = 0$ and $\ddot{y} + \frac{1}{2G}(G_v \dot{x}^2 + 2G_u \dot{x}\dot{y} - E_v \dot{y}^2) = 0$.
- **Vector Field along Curve:** A vector field X along a curve $c : I \rightarrow S$ is a map $X : I \rightarrow \mathbb{R}^3$ such that $X(t) \in T_{c(t)}S$ for all $t \in I$.
- **Covariant Derivative:** Let e_1, e_2, e_3 be a frame and $X(t) = f_1(t)e_1(t) + f_2(t)e_2(t)$ be a vector field along c . The covariant derivative of X along c is the vector field $\frac{\nabla}{dt}X$ along c defined by $\frac{\nabla}{dt}X(t) = \left(\dot{f}_1 + f_2 \omega_2^1(\dot{c}) \right) e_1(c(t)) + \left(\dot{f}_2 + f_1 \omega_1^2(\dot{c}) \right) e_2(c(t))$ for all $t \in I$.
- **Gauss-Weingarten Formula:** Let e_1, e_2, e_3 be an adapted frame on S , and $X(t)$ be a vector field along a curve $c : I \rightarrow S$. Then the derivative of X in $\mathbb{R}^3$ can be decomposed as $\frac{d}{dt}X(t) = \frac{\nabla}{dt}X(t) + II_{c(t)}(\dot{c}(t), X(t)) \cdot e_3(c(t))$.
- **Covariant Derivative and Geodesic:** Let c be a unit-speed curve and $\{e_1, e_2, e_3\}$ an adapted frame with $e_1(c(t)) = \dot{c}(t)$, then $\frac{\nabla}{dt}(\dot{c}(t)) = \kappa_g(t)e_2(c(t)), \kappa_n(t) = \kappa(\dot{c}(t))$.
Moreover, c is a geodesic $\iff \frac{\nabla}{dt}(\dot{c}(t)) = 0$.
- **Parallel Transport:** Given a curve $c : I \rightarrow S$ and a vector $X_0 \in T_{c(t_0)}S$, there exists a unique vector field X along c such that $X(0) = X_0$ and $\frac{\nabla}{dt}X(t) = 0$ for all $t \in I$. Such X is called the parallel transport of X_0 along c .
- **Holonomy Angle:** Given a curve $c : I \rightarrow S$ and the tangent frame $\{e_1, e_2\}$, the holonomy angle of the parallel transport along c is defined by $\text{Hol}(c) = - \int_c \omega_1^2$.

- **Energy of a Curve:** The energy of a curve $c : [a, b] \rightarrow S$ is $E(c) = \frac{1}{2} \int_a^b \|\dot{c}(t)\|^2 dt$.
- **Exponential Map:** Given $p \in S$ and $v \in T_p S$, there exists a unique geodesic $c_v : I \rightarrow S$ such that $c_v(0) = p$ and $\dot{c}_v(0) = v$. The exponential map at p is defined by $\exp_p : T_p S \rightarrow S, \exp_p(v) = c_v(1)$.

2.12 Fundamental Theorem of a Surface

- **Pullback of Fundamental Forms:** If S is a parameterized surface via $\psi : \mathbb{R}^2 \supseteq D \rightarrow S$, then $g = \begin{pmatrix} g_{11} & g_{12} \\ g_{12} & g_{22} \end{pmatrix} = \psi^* I, l = \begin{pmatrix} l_{11} & l_{12} \\ l_{12} & l_{22} \end{pmatrix} = \psi^* II$ where g and l is the families of symmetric metrics, parameterized by points $(x, y) \in D$.
- **Gauss-Codazzi Equations:** $d\omega_1^2 = \omega_1^3 \wedge \omega_3^2, d\omega_1^3 = \omega_1^2 \wedge \omega_2^3, d\omega_2^3 = \omega_2^1 \wedge \omega_1^3$.
- **Fundamental Theorem of a Surface:** Locally, a surface is uniquely determined by its first and second fundamental forms.

2.13 Gauss-Bonnet Theorem

- **Gauss-Bonnet Formula for Compact Oriented surface:** $\frac{1}{2\pi} \int_S K dA = 2(1 - \# \text{holes}) = \chi(S)$
- **Local version of Gauss-Bonnet Theorem:** Let S be a compact oriented surface with the area element dA . $\forall B \subseteq S$, whose boundary is smooth, admits an adapted, positive oriented frame, and ∂B is a simple, unit-speed curve $\partial B : [a, b] \rightarrow S$, we have $\int_B K dA + \int_a^b \kappa_g(t) dt = 2\pi$.
- **Euler Characteristic:** For a compact oriented surface S , the Euler characteristic is defined by $\chi(S) = V - E + F$. $\chi(\text{Sphere}) = 2, \chi(\text{Torus}) = 0, \chi(\text{Disc}) = 1$

3 Calculation Examples

3.1 Example: Sphere

Parametrization: $\Phi(\theta, \varphi) = (R \sin \theta \cos \varphi, R \sin \theta \sin \varphi, R \cos \theta)$.

Adapted Frame: $e_1 =$

- **Coframe:** $\omega^1 = R d\theta, \omega^2 = R \sin \theta d\varphi$.
- **Differentials:** $d\omega^1 = 0, d\omega^2 = R \cos \theta d\theta \wedge d\varphi = \frac{\cos \theta}{R \sin \theta} \omega^1 \wedge \omega^2$.
- **Connection Form:** Assume $\omega_{12} = A\omega^1 + B\omega^2$. $0 = d\omega^1 = \omega^2 \wedge (-\omega_{12}) \implies \omega_{12} \propto \omega^2$. $d\omega^2 = \omega^1 \wedge \omega_{12}$. Result: $\omega_{12} = -\cos \theta d\varphi$.
- **Curvature:** $d\omega_{12} = \sin \theta d\theta \wedge d\varphi = \frac{1}{R^2} \omega^1 \wedge \omega^2 \implies K = \frac{1}{R^2}, H = -\frac{1}{R}$.

3.1.1 Example: Cylinder

Parametrization: $\Phi(\theta, z) = (r \cos \theta, r \sin \theta, z)$.

- $E = r^2, F = 0, G = 1$.
- $e = -r, f = 0, g = 0$ (with inward normal).

- **Gaussian Curvature:** $K = \frac{(-r)(0) - 0}{r^2} = 0$.
- **Mean Curvature:** $H = -1/(2r)$.
- **Geodesics:** Assume $\gamma(t) = x(t), y(t)$, then $r^2\ddot{x} = 0, \ddot{y} = 0 \Rightarrow x(t) = at + b, y(t) = ct + d$.

3.1.2 Example: Torus

Parametrization: $\Phi(u, v) = ((R + r \cos v) \cos u, (R + r \cos v) \sin u, r \sin v)$.

- **Gaussian Curvature:** $K = \frac{\cos v}{r(R + r \cos v)}$
- Outer circle ($v = 0$): $K > 0$. Inner circle ($v = \pi$): $K < 0$. Top/Bottom ($v = \pm\pi/2$): $K = 0$.

3.1.3 Example: Surface of Revolution

Curve $c(t) = (r(t), 0, z(t))$ rotated around z-axis.

- **Gaussian Curvature:** $K = -\frac{r''(t)}{r(t)}$ (assuming unit speed profile)

3.2 Algorithm: Computing Curvature via Coframe

1. **Choose a Frame:** Find orthonormal e_1, e_2 tangent to S .
2. **Find Coframe:** Write $ds^2 = (\omega^1)^2 + (\omega^2)^2$. Identify ω^1, ω^2 .
3. **Differentiate:** Compute $d\omega^1$ and $d\omega^2$.
4. **Solve for ω_{12} :** Use the First Structure Equation:

$$d\omega^1 = \omega^2 \wedge \omega_{21}, \quad d\omega^2 = \omega^1 \wedge \omega_{12}$$

(Recall $\omega_{21} = -\omega_{12}$).

5. **Compute K :** Calculate $d\omega_{12}$ and write it as $-K\omega^1 \wedge \omega^2$.

3.3 The Darboux Frame (Frame along a Curve)

For a unit-speed curve $c(s)$ on a surface S :

- **Definition:** An orthonormal frame $\{T, V, n\}$ adapted to both the curve and the surface.

- $T(s) = \dot{c}(s)$ (Unit Tangent).
- $n(s)$ is the surface unit normal restricted to the curve.
- $S(s) = n(s) \times T(s)$ (Tangent Normal / Side Normal).

- **Darboux Equations**
$$\begin{pmatrix} T' \\ S' \\ n' \end{pmatrix} = \begin{pmatrix} 0 & -\kappa_g & -\kappa_n \\ \kappa_g & 0 & -\tau_g \\ \kappa_n & \tau_g & 0 \end{pmatrix} \begin{pmatrix} T \\ S \\ n \end{pmatrix}$$

- **Curvatures:**

- **Geodesic Curvature** (κ_g): Bending in the tangent plane. $\kappa_g = \langle T', V \rangle = \omega_{12}(T)$.
- **Normal Curvature** (κ_n): Bending normal to the surface. $\kappa_n = \langle T', n \rangle = II(T, T) = \omega_{13}(T)$.
- **Geodesic Torsion** (τ_g): Twisting of the normal. $\tau_g = \langle n', V \rangle = \omega_{23}(T)$.

- **Relation to Space Curvature:** The curvature vector of the curve is $\vec{\kappa} = T' = \kappa_g V + \kappa_n n$.

Squared curvature: $\kappa^2 = \kappa_g^2 + \kappa_n^2$.

4 Test Exam

4.1 EXERCISE 1. (True/False)

Are the following claims true or false? Give a short one sentence explanation for your answer.

- (i) For any given smooth functions $\kappa > 0$ and τ (defined on some interval) there exists a regular curve with curvature κ and torsion τ .

Answer: True. Curvature and torsion uniquely determine a curve.

- (ii) The subset of $\mathbb{R}^3$ defined by the equation $z = x^2 - y^2$ is a surface.

Answer: True. It can be seen as level set $g(x, y, z) = x^2 - y^2 - z = 0$. $\nabla g = (2x, -2y, -1) \neq 0$.

- (iii) Every surface admits a non-zero three-form.

Answer: False. If $n > m$, then a n -form on an m -dimensional manifold is always zero. Since a surface is 2-dimensional, any 3-form on it must be zero.

- (iv) The principal directions of a surface are intrinsic.

Answer: False. The principal direction is the eigenvector of the shape operator, and the definition of the shape operator depends on the choice of the normal vector n in the ambient space and the embedding, so the principal direction is an extrinsic geometric quantity.

- (v) If two isometric, connected surfaces in $\mathbb{R}^3$ have the same second fundamental form, then one is the image of the other under a composition of a translation, a rotation and a reflection.

Answer: True. The Fundamental Theorem of Surfaces gets this.

- (vi) If v is a tangent vector at a point p and S_p the shape operator, then $S_p(v)$ is a vector normal to the surface.

Answer: False.

- (vii) A curve c in $\mathbb{R}^3$ of constant speed is a geodesic if and only if the acceleration is tangent to the surface S at $c(t)$ for all t .

Answer: False. On the contrary, the definition of a geodesic is that the tangent vector field is parallel transported along the curve, which means the acceleration vector $\ddot{c}(t)$ has no tangent component and must be **normal** to the surface (i.e., $\ddot{c}(t) \parallel n$).

4.2 EXERCISE 2.

Let θ, φ be the standard spherical coordinates on S^2 , and consider the function $f : S^2 \rightarrow \mathbb{R}$ given by $f(x, y, z) = (x - y)^2 + z^2$. Calculate df in terms of θ, φ .

The parameterization of a unit sphere is given by $\Phi(\theta, \varphi) = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$.

Method 1: Then we have $f = (x - y)^2 + z^2 = x^2 - 2xy + y^2 + z^2 = (x^2 + y^2 + z^2) - 2xy$
 $= 1 - 2(\sin \theta \cos \varphi)(\sin \theta \sin \varphi)$
 $= 1 - 2 \sin^2 \theta \sin \varphi \cos \varphi = 1 - \sin^2 \theta \sin(2\varphi)$

Take the derivative, we get $df = \frac{\partial f}{\partial \theta} d\theta + \frac{\partial f}{\partial \varphi} d\varphi$
 $= (-2 \sin \theta \cos \theta \sin(2\varphi)) d\theta + (-\sin^2 \theta \cdot 2 \cos(2\varphi)) d\varphi$
 $= -\sin(2\theta) \sin(2\varphi) d\theta - 2 \sin^2 \theta \cos(2\varphi) d\varphi$

Method 2: Alternatively, we can compute df directly using the chain rule:

$$\begin{aligned}
 df &= \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz = 2(x-y)dx + 2(y-x)dy + 2zdz \\
 &= 2(\sin \theta \cos \varphi - \sin \theta \sin \varphi)(\cos \theta \cos \varphi d\theta - \sin \theta \sin \varphi d\varphi) \\
 &\quad + 2(\sin \theta \sin \varphi - \sin \theta \cos \varphi)(\cos \theta \sin \varphi d\theta + \sin \theta \cos \varphi d\varphi) + 2(\cos \theta)(-\sin \theta d\theta) \\
 &= 2 \sin \theta (\cos \varphi - \sin \varphi)(\cos \theta (\cos \varphi - \sin \varphi) d\theta - \sin \theta (\sin \varphi + \cos \varphi) d\varphi) - 2 \cos \theta \sin \theta d\theta \\
 &= -\sin(2\theta) \sin(2\varphi) d\theta - 2 \sin^2 \theta \cos(2\varphi) d\varphi
 \end{aligned}$$

4.3 EXERCISE 3.

The catenoid S is given by $\Psi(u, v) = (\cosh v \cos u, \cosh v \sin u, v)$, $u \in [0, 2\pi)$, $v \in \mathbb{R}$.

(i) **Determine an adapted moving frame $\{e_1, e_2, e_3\}$ on S .**

Take partial derivatives: $\Psi_u = (-\cosh v \sin u, \cosh v \cos u, 0)$, $\Psi_v = (\sinh v \cos u, \sinh v \sin u, 1)$.

These two vectors are orthogonal, thus we can unitize them to get the tangent frame e_1, e_2 .

$$e_1 = \frac{\Psi_u}{\|\Psi_u\|} = \frac{\Psi_u}{\cosh v} = (-\sin u, \cos u, 0), e_2 = \frac{\Psi_v}{\|\Psi_v\|} = \frac{\Psi_v}{\sqrt{\sinh^2 v + 1}} = \frac{\Psi_v}{\cosh v} = (\tanh v \cos u, \tanh v \sin u, \text{sech } v)$$

$$e_3 = e_1 \times e_2 \therefore e_3 = \begin{vmatrix} i & j & k \\ -\sin u & \cos u & 0 \\ \tanh v \cos u & \tanh v \sin u & \text{sech } v \end{vmatrix} = (\text{sech } v \cos u, \text{sech } v \sin u, -\tanh v)$$

(ii) **Calculate the dual frame $\{\omega^1, \omega^2\}$ and the connection forms ω_i^j .**

Since $e_1 = \frac{\partial_u}{\cosh v}$. And $\omega^1(e_1) = 1$, thus $\omega^1 = \cosh v du$. The same reason to the $\omega^2 = \cosh v dv$.

Another solution is $\begin{cases} dx = \omega^1 e_1 + \omega^2 e_2 \\ dx = \Phi_u du + \Phi_v dv = \cosh v e_1 du + \cosh v e_2 dv \end{cases}$ and we can get the same result.

$$\omega_1^2 = de_1 \cdot e_2 = (-\cos u du, -\sin u du) \cdot (\tanh v \cos u, \tanh v \sin u, \text{sech } v) = -\tanh v \cos^2 u du - \tanh v \sin^2 u du = -\tanh v du.$$

The same reason to get $\omega_1^3 = -\text{sech } v du$, $\omega_2^3 = -\text{sech } v dv$.

(iii) **Calculate the mean curvature H and the Gaussian curvature K of S .**

$$d\omega_1^2 = -K\omega^1 \wedge \omega^2 \text{ and } d\omega_1^2 = d(-\tanh v du) = -\text{sech}^2 v dv \wedge du = \text{sech}^2 v du \wedge dv, \omega^1 \wedge \omega^2 = \cosh^2 v du \wedge dv.$$

Compare the coefficients, we get $K = -\frac{1}{\cosh^4 v}$.

For mean curvature H , we have $\omega_{13} \wedge \omega^2 + \omega^1 \wedge \omega_{23} = 2H\omega^1 \wedge \omega^2$.

Substitute the values, we get $H = 0$.

(iv) **Verify that the total Gaussian curvature $\int_S K dA$ of S is $-4\pi \left(\frac{d}{dv} \tanh v = \frac{1}{\cosh^2 v} \right)$.**

$$dA = \omega^1 \wedge \omega^2 = \cosh^2 v du \wedge dv.$$

$$\begin{aligned}
 \text{Thus we have } \int_S K dA &= \int_{-\infty}^{\infty} \int_0^{2\pi} \left(-\frac{1}{\cosh^4 v} \right) (\cosh^2 v) du dv \\
 &= \int_{-\infty}^{\infty} \int_0^{2\pi} -\frac{1}{\cosh^2 v} du dv = 2\pi \int_{-\infty}^{\infty} -\text{sech}^2 v dv \\
 &= -2\pi [\tanh v]_{-\infty}^{\infty} = -2\pi(1 - (-1)) = -4\pi
 \end{aligned}$$

4.4 EXERCISE 4.

Assume the first fundamental form of a surface S in a certain parametrization is given by $I_{ij} = \begin{pmatrix} 1/y^2 & 0 \\ 0 & 1/y^2 \end{pmatrix}$ on the half-plane $D = \{y > 0\}$.

(i) **Calculate a tangent frame field e_1, e_2 on D .**

Using the first fundamental form, we get $\langle \partial_x, \partial_x \rangle = \frac{1}{y^2}$, thus $\|\partial_x\| = \frac{1}{y}$, so we have $\mathbf{e}_1 = y\partial_x$. Similarly, we get $\mathbf{e}_2 = y\partial_y$.

(ii) **Determine the connection form ω_1^2 uniquely defined by the structure equation.**

Firstly, we can get the dual frame $\omega^1 = \frac{1}{y}dx, \omega^2 = \frac{1}{y}dy$.

Assume $\omega_1^2 = Adu + Bdv$, then we have $d\omega^1 = \omega_1^2 \wedge \omega^2$ and $d\omega^2 = \omega^1 \wedge \omega_1^2$.
 $d\omega^1 = -\frac{1}{y^2}dy \wedge dx = \frac{1}{y^2}dx \wedge dy = (Adu + Bdv) \wedge \frac{1}{y}dy = \frac{A}{y}du \wedge dy \Rightarrow A = \frac{1}{y}$.
 $d\omega^2 = -\frac{1}{y^2}dy \wedge dy = 0 = (\frac{1}{y}dx) \wedge (Adu + Bdv) = \frac{B}{y}du \wedge dy \Rightarrow B = 0$.

Thus we have $\omega_1^2 = \frac{1}{y}dx$.

(iii) **Determine the geodesics.**

The first fundamental form is $I_{ij} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} = \begin{pmatrix} 1/y^2 & 0 \\ 0 & 1/y^2 \end{pmatrix}$. We know that a regular curve $\gamma(t) = (x(t), y(t))$ on the surface is a geodesic if it satisfies the following system of differential equations: $\ddot{x} + \frac{1}{2E}(E_x\dot{x}^2 + 2E_y\dot{x}\dot{y} - G_x\dot{y}^2) = 0, \ddot{y} + \frac{1}{2G}(G_y\dot{x}^2 + 2G_x\dot{x}\dot{y} - E_y\dot{y}^2) = 0$.

Given $E = y^{-2}$, $G = y^{-2}$, and $F = 0$, the partial derivatives are: $E_x = \frac{\partial}{\partial x}(y^{-2}) = 0, E_y = \frac{\partial}{\partial y}(y^{-2}) = -2y^{-3}, G_x = \frac{\partial}{\partial x}(y^{-2}) = 0, G_y = \frac{\partial}{\partial y}(y^{-2}) = -2y^{-3}$.

Substituting into the first equation, we have $\ddot{x} + \frac{y^2}{2}(0 \cdot \dot{x}^2 + 2(-2y^{-3})\dot{x}\dot{y} - 0 \cdot \dot{y}^2) = \ddot{x} - \frac{2\dot{x}\dot{y}}{y} = 0$.

And the same to the second equation, $\ddot{y} + \frac{y^2}{2}((-2y^{-3})\dot{x}^2 + 2(0)\dot{x}\dot{y} - (-2y^{-3})\dot{y}^2) = \ddot{y} - \frac{1}{y}(\dot{x}^2 - \dot{y}^2) = 0$.

Case 1: $\dot{x} = 0$, then $x(t) = x_0, \ddot{x} = 0$, therefore $\ddot{y} + \frac{1}{y}\dot{y}^2 = 0 \Rightarrow y = \sqrt{At + B}$.

Case 2: $\dot{x} \neq 0$, from the first equation, we have $\frac{\ddot{x}}{\dot{x}} = \frac{2\dot{y}}{y}$, integrating both sides with respect to t , we get $\ln|\dot{x}| = 2\ln|y| + C_1 \Rightarrow \dot{x} = cy^2$.

Since the geodesic has constant speed, we can assume unit speed without loss of generality.

The length element squared is $ds^2 = \frac{dx^2 + dy^2}{y^2} = 1$. Thus we have $\frac{\dot{x}^2 + \dot{y}^2}{y^2} = 1 \Rightarrow \dot{x}^2 + \dot{y}^2 = y^2$.

Substitute $\dot{x} = cy^2$ into the speed equation, we have $(cy^2)^2 + \dot{y}^2 = y^2 \Rightarrow c^2y^4 + \dot{y}^2 = y^2 \Rightarrow \dot{y}^2 = y^2(1 - c^2y^2)$.

Thus $\frac{dy}{dx} = y^2(1 - c^2y^2) \Rightarrow \pm dx = \frac{cy}{\sqrt{1 - c^2y^2}} dy$.

Take the integral, $\int \frac{cy}{\sqrt{1-c^2y^2}} dy = \frac{-1}{2c} \int \frac{1}{\sqrt{1-c^2y^2}} d(1-c^2y^2) = -\frac{1}{2c} \int \frac{1}{u^{-\frac{1}{2}}} du = -\frac{1}{c} \sqrt{1-c^2y^2}.$

Therefore, we have $-\frac{1}{c} \sqrt{1-c^2y^2} = \pm(x-x_0) \Rightarrow (\mathbf{x}-\mathbf{x}_0)^2 + \mathbf{y}^2 = \left(\frac{1}{c}\right)^2$

4.5 EXERCISE 5.

Curve $c(t) = \left(t, \frac{1+t}{t}, \frac{1-t^2}{t}\right)$ for $t \in (0, \infty)$.

(i) **Prove that c is regular.**

$\dot{c}(t) = (1, -t^{-2}, -t^{-2} - 1)$. The first component is always 1, thus $\|\dot{c}(t)\| \neq 0$ for all t .

(ii) **Compute the torsion of c .**

$$\ddot{c}(t) = (0, 2t^{-3}, 2t^{-3}). \text{ Then } \dot{c} \times \ddot{c} = \begin{vmatrix} i & j & k \\ 1 & -t^{-2} & -1-t^{-2} \\ 0 & 2t^{-3} & 2t^{-3} \end{vmatrix} = 2t^{-3}(1, -1, 1).$$

Therefore, the curvature κ is given by: $\kappa = \frac{\|\dot{c} \times \ddot{c}\|}{\|\dot{c}\|^3} = \frac{2\sqrt{3}/t^3}{\left(\frac{\sqrt{2(t^4+t^2+1)}}{t^2}\right)^3} = \frac{\sqrt{6}t^3}{(t^4+t^2+1)^{3/2}}$

(iii) **Compute the torsion of c .**

$$\ddot{c} = (0, -6t^{-4}, -6t^{-4}). (\dot{c} \times \ddot{c}) \cdot \ddot{c} = 2t^{-3}(1, -1, 1) \cdot -6t^{-4}(0, 1, 1) = -12t^{-7}(-1+1) = 0.$$

$$\text{Thus, the torsion } \tau \text{ is } \tau = \frac{(\dot{c} \times \ddot{c}) \cdot \ddot{c}}{\|\dot{c} \times \ddot{c}\|^2} = \frac{0}{\|2t^{-3}(1, -1, 1)\|^2} = 0.$$

4.6 EXERCISE 6.

Let S be a compact surface with unit normal field $n : S \rightarrow S^2$.

(i) **Prove that n is surjective onto S^2 .**

Let $v \in S^2$ be an arbitrary unit vector in $\mathbb{R}^3$. We want to show that there exists a point $p \in S$ such that $n(p) = v$.

Consider the *height function* $h_v : S \rightarrow \mathbb{R}$ defined by the dot product: $h_v(x) = x \cdot v$, for all $x \in S$.

Since S is compact and h_v is continuous, h_v must achieve its global maximum at some point $p \in S$.

Let p be a point where h_v attains its maximum. At a critical point of a function defined on a surface, the gradient of the restriction is zero. Geometrically, this means the tangent plane $T_p S$ is orthogonal to the gradient of the ambient linear function $f(x) = x \cdot v$. The gradient of f is simply the vector v . Therefore, the vector v is normal to the tangent plane $T_p S$.

This implies that the unit normal $n(p)$ is parallel to v . That is: $n(p) = v$ or $n(p) = -v$.

To distinguish between these two, observe that p is a global maximum. This means that for all $x \in S$, $x \cdot v \leq p \cdot v$. Geometrically, the entire surface S lies in the half-space defined by the plane passing through p with normal v .

Assuming the standard convention that n is the *outward* pointing normal of a compact surface, the normal at the "highest" point in the direction v must point in the direction of v (out of the bounded region enclosed by S). Therefore, $n(p) = v$.

Since $v \in S^2$ was arbitrary, for every unit vector v , there exists a point $p \in S$ such that $n(p) = v$. Thus, n is surjective.

- (ii) **Let $K^+(p) = \max\{0, K(p)\}$. Show that $\int_S K^+ dA \geq 4\pi$.**

Since n is surjective, for every unit vector $v \in S^2$, there exists at least one point $p \in S$ such that $n(p) = v$. Each such point p corresponds to a global maximum of the height function in the direction of v , and at these points, the Gaussian curvature $K(p)$ is non-negative. By considering the area of the sphere and the contributions from these points, we can show that the total positive curvature must be at least 4π .

- (iii) **Determine the image of n for the catenoid and the cylinder.**

For the Catenoid, the normal vector is $\mathbf{n} = (\text{sech } v \cos u, \text{sech } v \sin u, -\tanh v)$. It covers the entire sphere except the north and south poles as v varies over $(-\infty, \infty)$.

For the Cylinder, the normal vector is $\mathbf{n} = (\cos u, \sin u, 0)$. It only covers the equator of the sphere as u varies over $[0, 2\pi)$.

4.7 EXERCISE 7.

Compact, oriented surface S with mean curvature H and Gauss curvature K .

- (i) **Prove that $H^2 \geq K$.**

By the definition, $H = \frac{\kappa_1 + \kappa_2}{2}$, $K = \kappa_1 \kappa_2$. $H^2 - K = \left(\frac{\kappa_1 + \kappa_2}{2}\right)^2 - \kappa_1 \kappa_2 = \frac{(\kappa_1 - \kappa_2)^2}{2} \geq 0$.

- (ii) **Show that S cannot be a minimal surface.**

If S were a minimal surface, then $H = 0$ for all points on S . From (i), we have $0 = H^2 \geq K \Rightarrow K \leq 0$ for all points on S .

But S is a compact surface, there exists at least one point p such that $K(p) > 0$. This contradicts the previous result that $K \leq 0$ everywhere on S .

Therefore, S cannot be a minimal surface.

- (iii) **Assume S is not the sphere and that it does not have a smooth bijection to a sphere. Prove that then there are points on S where the Gaussian curvature is positive, negative and zero.**

Since S is compact, there exists at least one point p such that $K(p) > 0$.

By the Gauss-Bonnet theorem, $\int_S K dA = 2\pi \chi(S)$. Since S is not a sphere, its Euler characteristic $\chi(S) \leq 0$. Thus, the total Gaussian curvature $\int_S K dA \leq 0$. Thus, there must be points on S where $K < 0$ to ensure that the total curvature is non-positive.

Since S is connected and K is a continuous function, by the Intermediate Value Theorem, there must be points on S where $K = 0$ as well.

5 Appendix

5.1 Integrals of Special Functions

$+C$ and dx is omitted in the following formulas for brevity.

Trigonometric Functions

- $\int \tan x = -\ln |\cos x|$
- $\int \cot x = \ln |\sin x|$
- $\int \sec x = \ln |\sec x + \tan x|$
- $\int \csc x = \ln |\csc x - \cot x|$

Inverse Trigonometric Functions

- $\int \arcsin x = x \arcsin x + \sqrt{1 - x^2}$
- $\int \arccos x = x \arccos x - \sqrt{1 - x^2}$
- $\int \arctan x = x \arctan x - \frac{1}{2} \ln(1 + x^2)$
- $\int \operatorname{arccot} x = x \operatorname{arccot} x + \frac{1}{2} \ln(1 + x^2)$
- $\int \operatorname{arcsec} x = x \operatorname{arcsec} x - \ln |x + \sqrt{x^2 - 1}|$
- $\int \operatorname{arccsc} x = x \operatorname{arccsc} x + \ln |x + \sqrt{x^2 - 1}|$

Hyperbolic Functions

- $\int \sinh x = \cosh x$
- $\int \cosh x = \sinh x$
- $\int \tanh x = \ln(\cosh x)$
- $\int \coth x = \ln |\sinh x|$
- $\int \operatorname{sech} x = \arctan(\sinh x)$
- $\int \operatorname{csch} x = \ln \left| \tanh \frac{x}{2} \right|$

Inverse Hyperbolic Functions

- $\int \operatorname{arsinh} x = x \operatorname{arsinh} x - \sqrt{x^2 + 1}$
- $\int \operatorname{arcosh} x = x \operatorname{arcosh} x - \sqrt{x^2 - 1}$
- $\int \operatorname{artanh} x = x \operatorname{artanh} x + \frac{1}{2} \ln(1 - x^2)$
- $\int \operatorname{arcoth} x = x \operatorname{arcoth} x + \frac{1}{2} \ln(x^2 - 1)$
- $\int \operatorname{arsech} x = x \operatorname{arsech} x + \arcsin x$
- $\int \operatorname{arcsch} x = x \operatorname{arcsch} x + \operatorname{arsinh} x$

Rational/Irrational Integrals

- $\int \frac{1}{\sqrt{x^2 + a^2}} = \ln |x + \sqrt{x^2 + a^2}|$
- $\int \frac{1}{\sqrt{x^2 - a^2}} = \ln |x + \sqrt{x^2 - a^2}|$
- $\int \frac{1}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{|a|}$
- $\int \frac{1}{x^2 + a^2} = \frac{1}{|a|} \arctan \frac{x}{|a|}$
- $\int \frac{1}{x^2 - a^2} = \frac{1}{2|a|} \ln \left| \frac{x - a}{x + a} \right|$
- $\int \sqrt{x^2 + a^2} = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ln |x + \sqrt{x^2 + a^2}|$
- $\int \sqrt{x^2 - a^2} = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln |x + \sqrt{x^2 - a^2}|$
- $\int \sqrt{a^2 - x^2} = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin \frac{x}{|a|}$
- $\int \frac{1}{\sqrt{(x^2 + a^2)^3}} = \frac{x}{a^2 \sqrt{x^2 + a^2}}$
- $\int \frac{1}{\sqrt{(x^2 - a^2)^3}} = -\frac{x}{a^2 \sqrt{x^2 - a^2}}$

5.2 Hyperbolic Trigonometric Identities

- $\sinh x = \frac{e^x - e^{-x}}{2}, \cosh x = \frac{e^x + e^{-x}}{2}$
- $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- $\cosh^2 x - \sinh^2 x = 1$

- $\operatorname{sech}^2 x + \tanh^2 x = 1$
- $\coth^2 x - \operatorname{csch}^2 x = 1$
- $\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$
- $\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$
- $\tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$
- $\operatorname{arcsinh} x = \ln x + \sqrt{x^2 + 1}$
- $\operatorname{arcosh} x = \ln x + \sqrt{x^2 - 1}$
- $\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$

5.3 Translations of common Curves and Surfaces

- Cylinder: 圆柱面
- Torus: 环面
- Catenoid: 双曲面 (悬链面)
- Helicoid: 螺旋面
- Ellipsoid: 椭球面
- elliptic paraboloid: 椭圆抛物面
- hyperbolic paraboloid: 双曲抛物面
- hyperboloid of one sheet: 单叶双曲面
- hyperboloid of two sheets: 双叶双曲面
- Saddle surface: 马鞍面
- Cone: 锥面
- Helix: 螺旋线
- Cycloid: 摆线
- Ellipse: 椭圆
- Parabola: 抛物线
- Hyperbola: 双曲线